

Evaluating Machine Learning Models for Predicting Commuting Route Disruptions in London

Table of Contents

Introduction2

Dataset Description.....3

 1. Commute Records Dataset.....3

 2. User Profile Dataset.....4

Exploratory Data Analysis (EDA) Findings.....5

 Feature Correlation Matrix for Commuting Route Disruption Prediction.....5

 Categorical Variables5

 Analysis of Commute Punctuality and Delay Proportions.....6

 Analysis of Factors Influencing Commuting Duration.....7

Feature Engineering and Model Implementation.....9

 Feature Engineering9

 Model Implementation9

Model Evaluation and Performance..... 11

 Validation method 11

 Performance metrics..... 11

 Evaluation..... 12

Discussion and Interpretation 13

Conclusion and Future Directions 14

Bibliography 15

Group Presentation Link..... 15

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Introduction

In cities that rely on public transport, delays and disruptions can affect people's daily routines and work. In London, where the underground and rail network are key, these issues lead to lost time. As a result, predicting disruption risk in advance is an important aim in intelligent transport systems. This project builds an AI-Enhanced Route Forecaster to predict daily disruption risk. It uses past commuting data, time patterns, route details, and weather to identify risk factors and give early warnings for better journey planning. Machine learning models are trained to support travel planning and traffic management.

Dataset Description

This project utilises two complementary datasets: **Commute Records** and **User Profiles**. These datasets are merged using the `User_id` field to create a comprehensive view of commuting behaviours and preferences.

1. Commute Records Dataset

This dataset has 1,500 rows and 13 columns, with each row showing one user's commute on a given day. Each columns cover specific features, such as commuting behaviour, time and location factors, route details, and weather. The data is synthetic and was created based on the operating features of London's public transport system to support the prediction of possible disruption. Below is an overview of the dataset's columns:

1. **User_id**: A unique identifier for commuting users.
2. **Date**: The specific date on which the commute occurred.
3. **Day_of_week**: The day of the week corresponding to the commute date.
4. **Hour_of_day**: The hour (0–23) when the commute commenced.
5. **Is_peak_hour**: Whether the commute occurred during peak hours (1 = peak period, 0 = off-peak period).
6. **Start_zone**: The commuting origin zone.
7. **End_zone**: The destination zone.
8. **Distance_km**: Estimated commute distance (in kilometres).
9. **Is_rainy**: Whether it rained on the commute day (1 = rainfall, 0 = no rainfall).
10. **Avg_trip_time_min**: Average commute duration (in minutes).
11. **Line_id**: Name of the public transport lines used for commute.
12. **Historical_delay_rate**: A numerical indicator of a line's historical delay rate.
13. **Route_status**: Whether the commute experienced route disruption or significant delay (1 = disruption occurred, 0 = normal operation).

2. User Profiles Dataset

This dataset contains 30 rows and 6 columns, representing static profiles for each of the 30 unique users tracked in the commute records. It provides context about the users' typical commuting preferences and allows for feature engineering.

1. **User_id**: Unique identifier to link with the Commute Records.
2. **Home_zone**: The user's residential zone.
3. **Work_zone**: The user's workplace zone.
4. **Primary_line**: The public transport line the user prefers to take.
5. **Backup_line**: An alternative line the user might take.
6. **Usual_hour**: The typical hour of day the user starts their commute.

Exploratory Data Analysis (EDA) Findings

Feature Correlation Matrix for Commuting Route Disruption Prediction

The heatmap shows a strong link between Distance_km and Avg_trip_time_min. Is_peak_hour has only weak to medium relationship with Route_status, and most other features are weak or near-zero, which suggests delays come from several factors together.

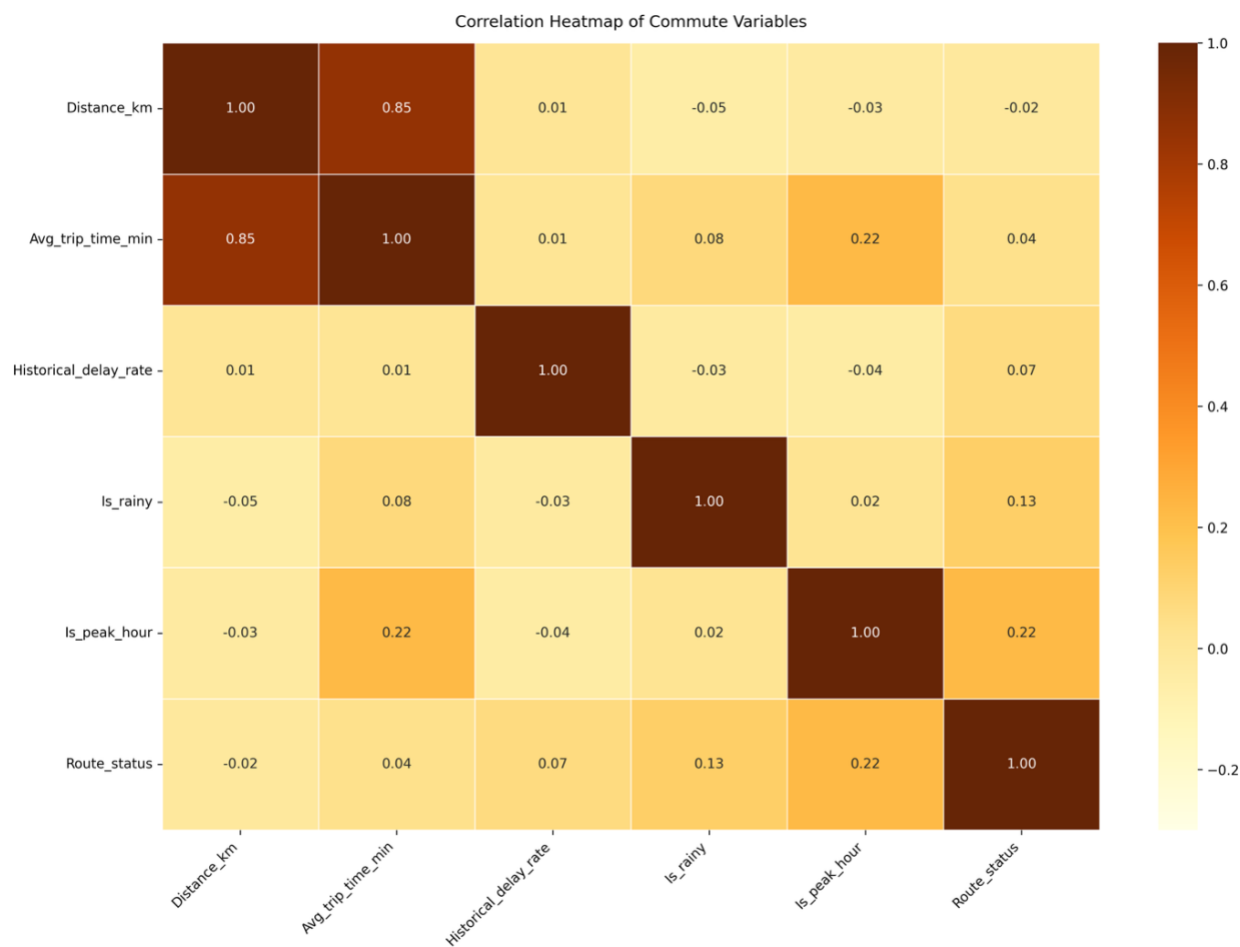


Figure 1 - Feature Correlation Matrix

Categorical Variables

These charts show how the main categorical variables are spread across the dataset, with each sub-chart showing the number of journeys in each category. Overall, journeys are spread fairly evenly across the week. The Central, Jubilee, and Northern lines appear most

often. Most journeys are recorded as on time, while delays make up a smaller share. For origin zones, Zone 2 have slightly more journeys than other zones, reflecting how many commuters travel from areas close to the city centre.

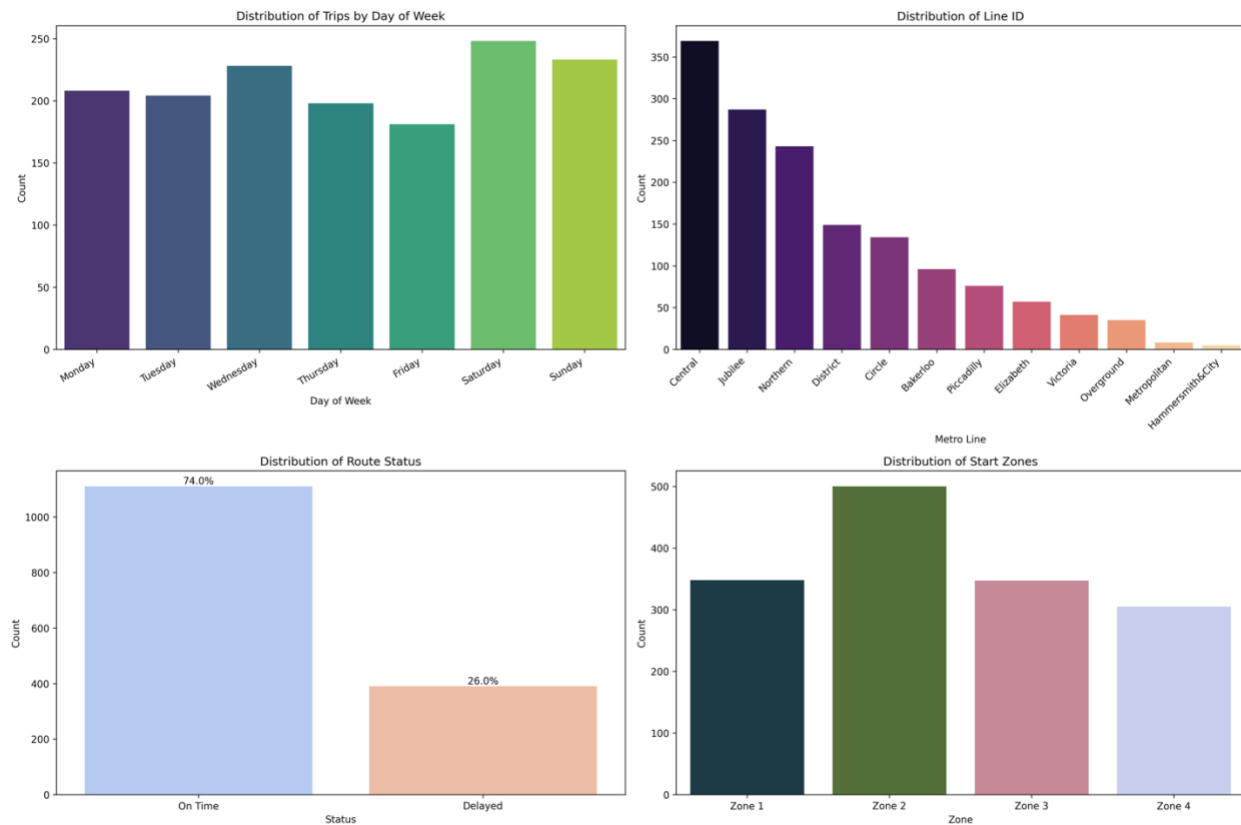


Figure 2 - Categorical Variables Analysis

Analysis of Commute Punctuality and Delay Proportions

The pie chart shows how many commutes were on time compared with delayed across the routes. It indicates that disruptions are not very common but still important.

Commute Reliability: On Time vs. Delayed

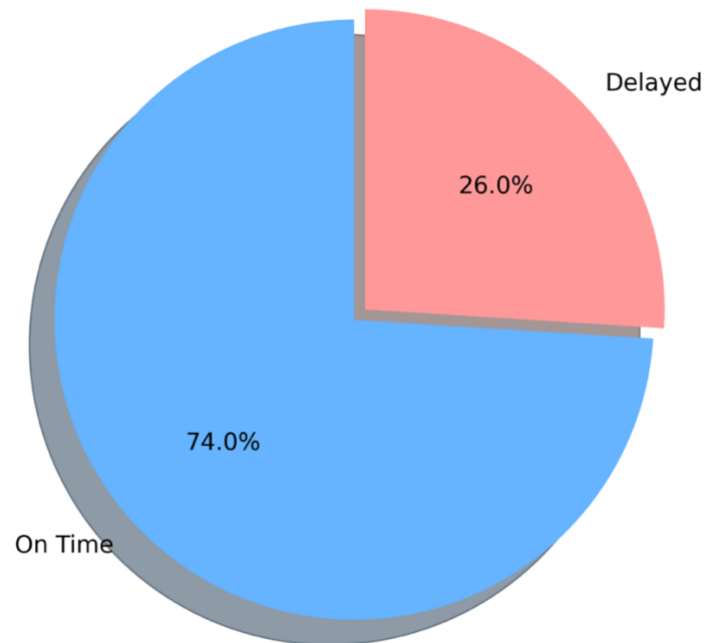


Figure 3 - Commute Reliability

Analysis of Factors Influencing Commuting Duration

The box plots below indicate that commuting times differ significantly across different lines and starting areas, while the distribution of commuting times across different days of the week is similar overall. This shows that line and zone factors are stronger determinants of trip duration than the specific day of the week.

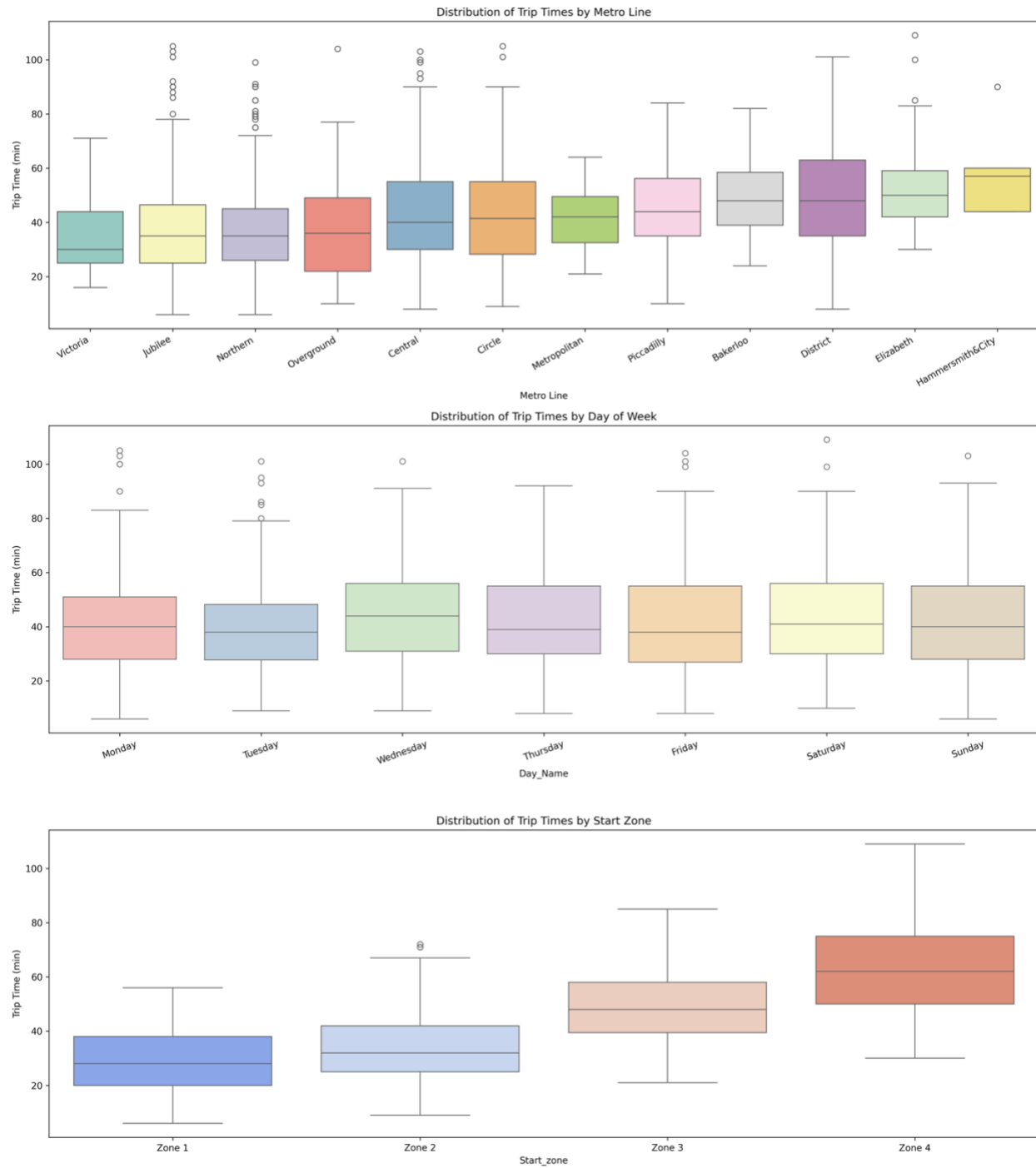


Figure 4 - Distribution of Trip Times

Feature Engineering and Model Implementation

Feature Engineering

Feature Selection: Some variables were removed because they were not useful for predicting delays. User_id and Date were dropped as they are only identifiers, and data-generation fields were removed to reduce overfitting and avoid data leakage.

Encoding Categorical Data: Categorical variables were converted using one-hot encoding, which treats categories separately and avoids a false order between categories, for example, implying Zone 2 is “higher than” Zone 1.

Handling Missing Values: No missing values were found in the dataset. We also followed a general rule: we removed any rows with missing category values and filled in missing number values using the mean.

Feature Transformation: Numerical features were standardised, so that they have similar scales. This helps prevent any single feature from having too much influence on the model and improves training stability.

Balancing the Dataset: The dataset is imbalanced, with more on-time journeys than delayed ones. To address this, we used SMOTE to increase the number of delayed samples. This method creates new samples based on nearby existing ones rather than copying old data, which helps reduce overfitting and improve model performance.

Model Implementation

Four supervised learning algorithms were selected to predict route disruptions: Decision Tree, Random Forest, XGBoost, and Logistic Regression.

Decision Tree: A decision tree is a rule-based model that splits data into branches based on feature values. Each split is a decision, and each leaf node gives a prediction on whether a route is disrupted or not (Analytix Labs, 2022).

Random Forest: Random Forest is an ensemble method that builds many decision trees separately and then combines their results to make a final prediction. This helps reduce overfitting and makes the model more stable than using just one tree (Kavlakoglu, no date).

XGBoost: XGBoost is a boosting-based ensemble algorithm that builds decision trees sequentially, where each new tree focuses on correcting the errors made by previous one. It was included to evaluate whether a more advanced model could improve prediction accuracy (Bentéjac et al., 2019).

Logistic Regression: Logistic regression is a statistical model, which is commonly used for binary classification problems (Scikit-learn, no date). It estimates the probability of route disruption by applying a sigmoid function to a linear combination of input features. The model is simple, easy to implement, and interpretable (Baladram, 2024).

Model Evaluation and Performance

Validation method

To assess how well the models perform on unseen data, the data set was split into 3 parts:

- Training: 70%
- Validation: 15%
- Testing: 15%

Performance metrics

Model	Accuracy	Precision	Recall	F1 Score	AUC-ROC
Decision Tree	0.7087	0.7299	0.7175	0.7236	0.7081
Random Forest	0.7748	0.8000	0.7684	0.7839	0.8760
Logistic Regression	0.6366	0.6489	0.6893	0.6685	0.7106
XGBoost	0.7598	0.8050	0.7232	0.7619	0.8561

Table 1 - Performance Metrics

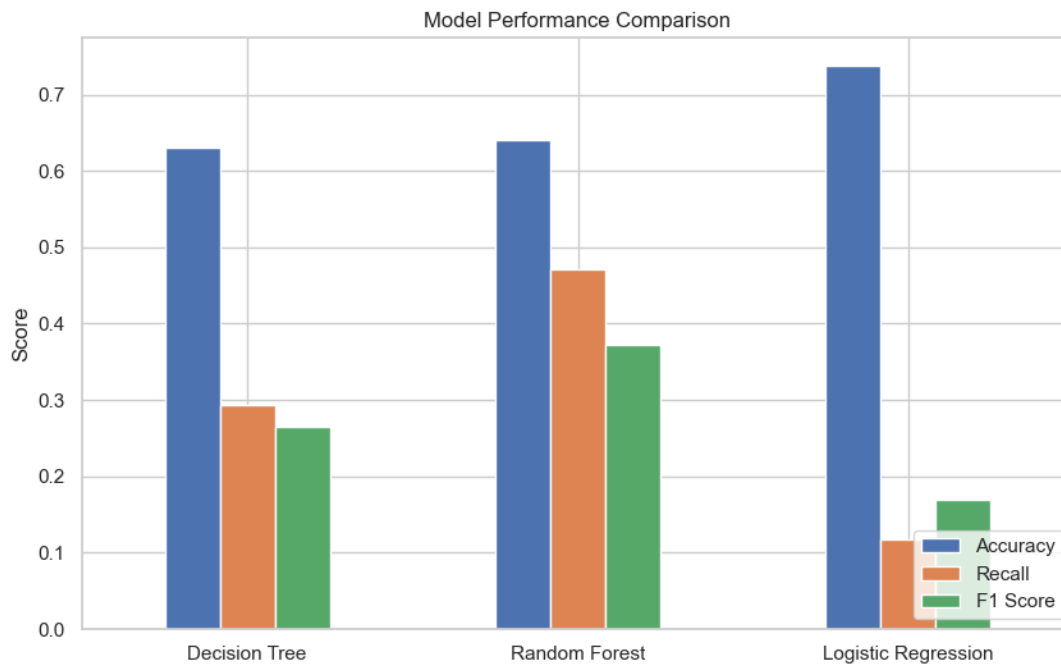


Figure 5 - Model Performance Comparison

Evaluation

Recall is treated as the most important metric in evaluation stage. In commuting, a False Negative (predicting “On Time” when a delay happens) is worse than a False Positive (predicting “Delay” when it is actually on time), as it can leave users unprepared and late. However, Accuracy and F1 score are also important to keep the system balanced overall.

Random Forest: Random Forest performed best overall. It achieved the highest Accuracy (0.7748), Recall (0.7684), F1 Score (0.7839), and AUC-ROC (0.8760), means it can clearly tell disrupted routes from normal ones. This is likely because it can learn complex non-linear patterns while being less prone to overfitting than a single Decision Tree.

XGBoost: XGBoost performed well and was the second-best model. It achieved the highest Precision (0.8050), and also gave fewer false delay warnings than the others. However, its Recall (0.7232) was a bit lower than Random Forest, so it missed slightly more real delays in this setup.

Decision Tree: The Decision Tree provided a reasonable baseline with an Accuracy of 0.7087. While simple and interpretable, it underperformed compared to the ensemble methods above, likely due to its tendency to overfit the training data and inability to generalise as well to unseen patterns.

Logistic Regression: Logistic Regression had the lowest performance across all metrics. This suggests that the relationship between the features and the route status is non-linear and complex, which the linear logistic model failed to capture effectively.

Discussion and Interpretation

The results suggests that ensemble models work well for predicting transport delays. Random Forest performed best overall, and combining strong delay detection with high accuracy. XGBoost also performed well, with particularly high Precision, means its delay predictions are usually dependable. Feature importance from tree-based models highlights key predictors such as Line_id, Hour_of_day, and Is_peak_hour, which reflect line-related delay trends and heavier disruption during peak times. The weaker performance of Logistic Regression shows that simple linear relationships are not enough, as delays depend on more complex conditions, such as rain affecting some lines more than others. Tree-based models can learn these kinds of conditional patterns more naturally.

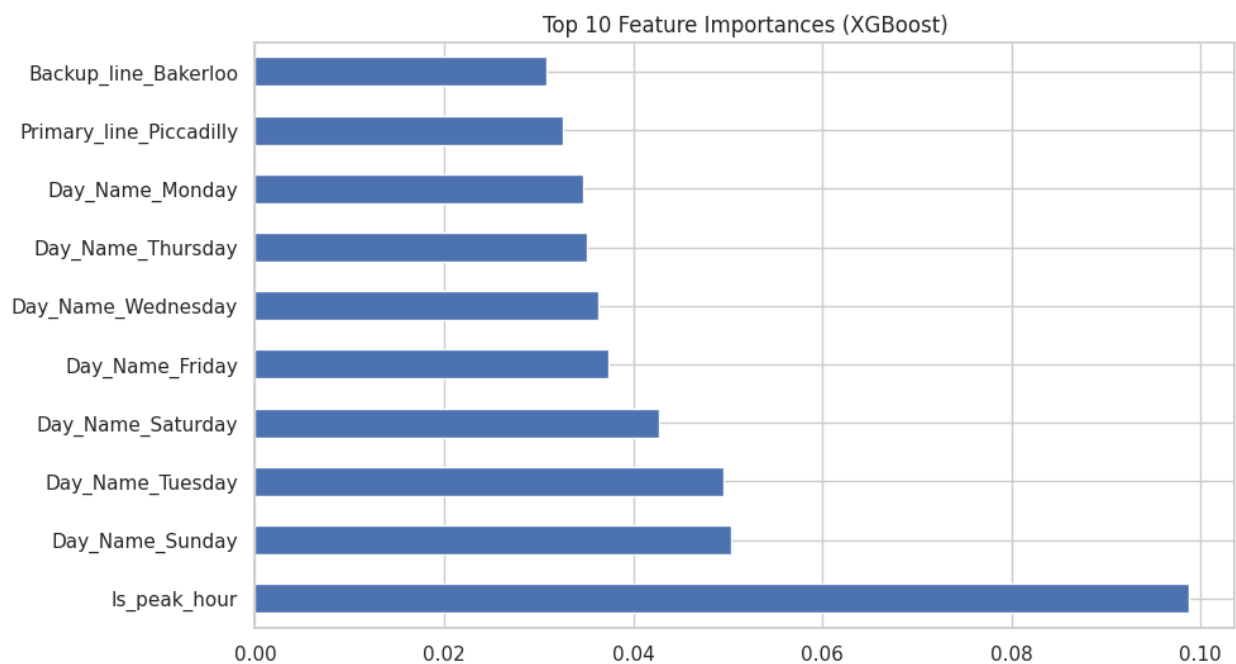


Figure 6 - Feature Importance

Conclusion and Future Directions

This project successfully built predictive models to forecast commuting route disruptions, with Random Forest achieving the best overall results. The findings show that potential delays can be predicted with reasonable accuracy using relevant factors. We indicate the possible next steps include:

1. Real-time data and Social Media signals integration: Add live APIs data to capture issues like signal failures or accidents, and also use social media text analysis to detect early reports of delays before they appear in official updates.
2. More detailed weather data: Replace the simple “Rainy” label with hourly rainfall levels, to better measure weather effects.
3. Deep learning: Test LSTM models to better learn time patterns in sequence and time-series data.

These improvements would help move the system from a static, history-based tool to a more dynamic, real-time travel assistant.

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Group Presentation Link

https://drive.google.com/file/d/1pLvoYkzJxuer3-NjOGhPIEq-EEpEV4BU/view?usp=drive_link